FROM RESEARCH TO INDUSTRY

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REQUIREMENTS ANALYSIS AND SYSTEM BOUNDS DEFINITION

Introduction on UML for Industrial Systems

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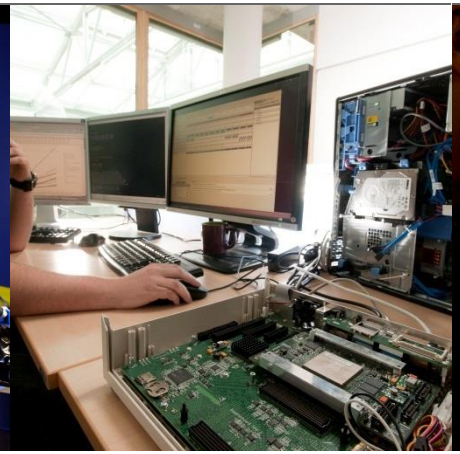

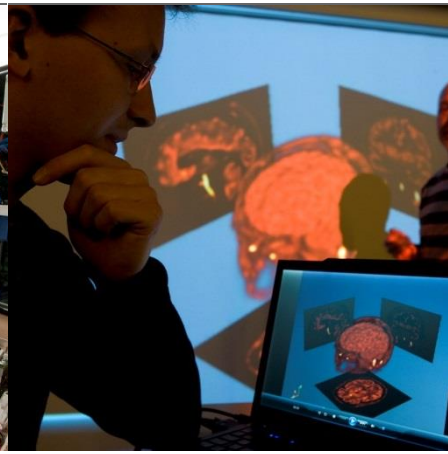

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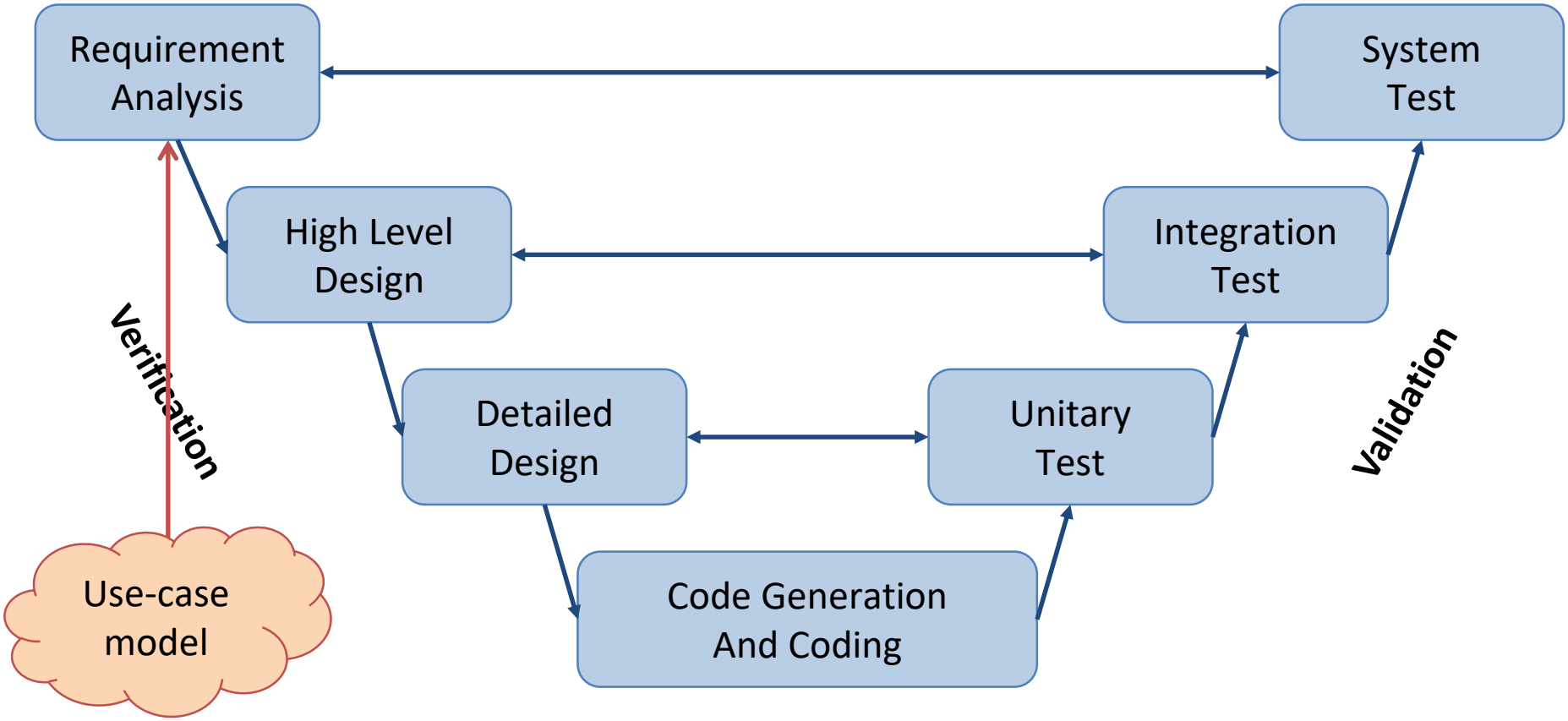
What are requirements and why model them?

- **Before developing, it is important to know:**
  - What the system does
  - What kind of needs it fulfills
- **These are the requirements of the system that you can model in a languages like SysML with dedicated stereotypes**
- **Some requirements can be functional:**
  - List functionalities offered by the system
  - Organize functionalities between them, so to represent the relationships between them
  - Functionalities can be re-used

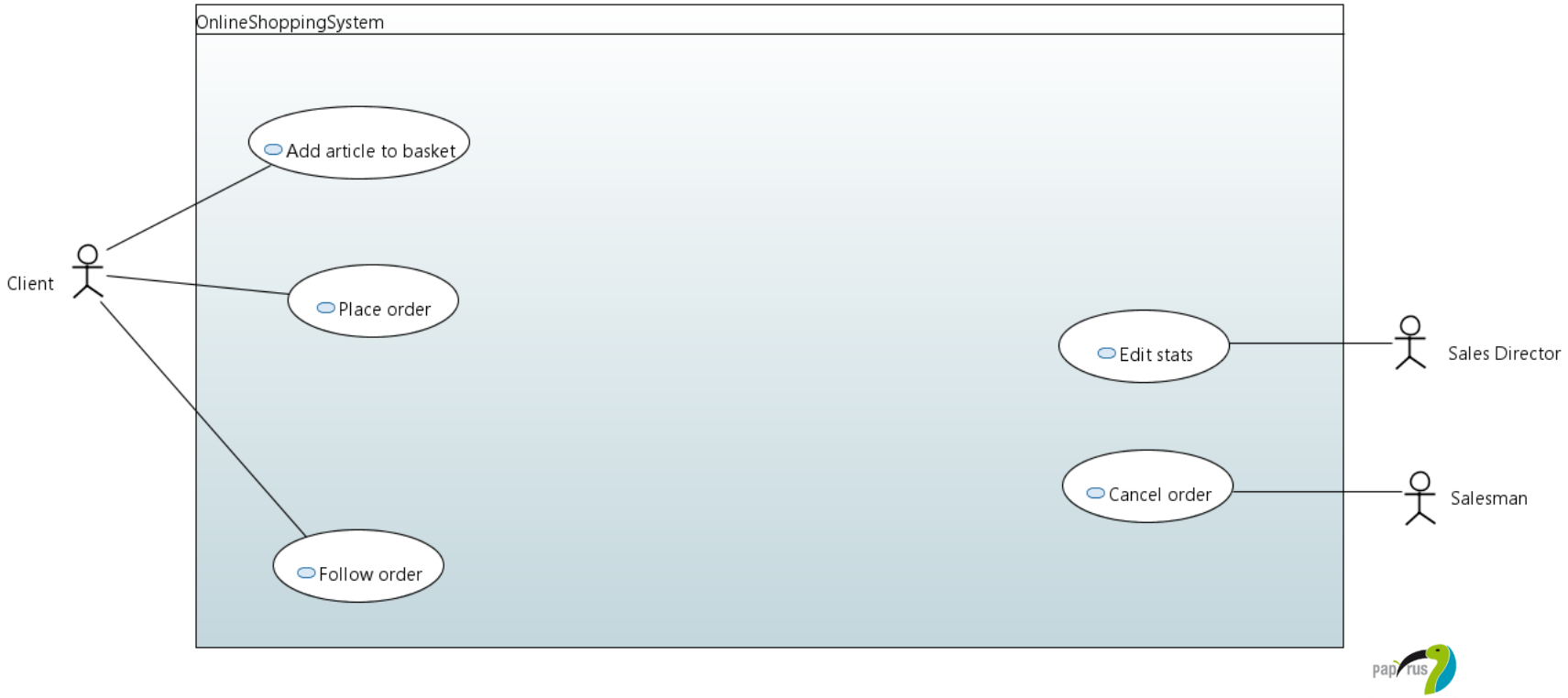
Functional requirements modeling in UML

- **Functional requirements can be modeled in UML with use-case diagrams**

When to model use-cases?



Example of simple use-case model



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BASIC ELEMENTS OF USE-CASE MODEL

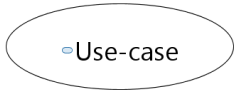
Subject

- The system that offers functionalities, seen as a black box
- Can be modeled with a class in UML

Reminder: Class = set of objects with same features, constraints and semantics

Use-case

- A functionality offered by the system
- Implies a series of elemental actions to be executed
- Example:

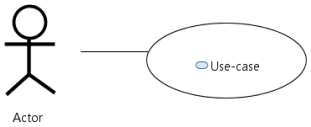


Actor

- Exterior entity to the system which interacts with the system
- Example:

Relationship between actor and use-case

- Actor is related with use-case through an association
- Example:



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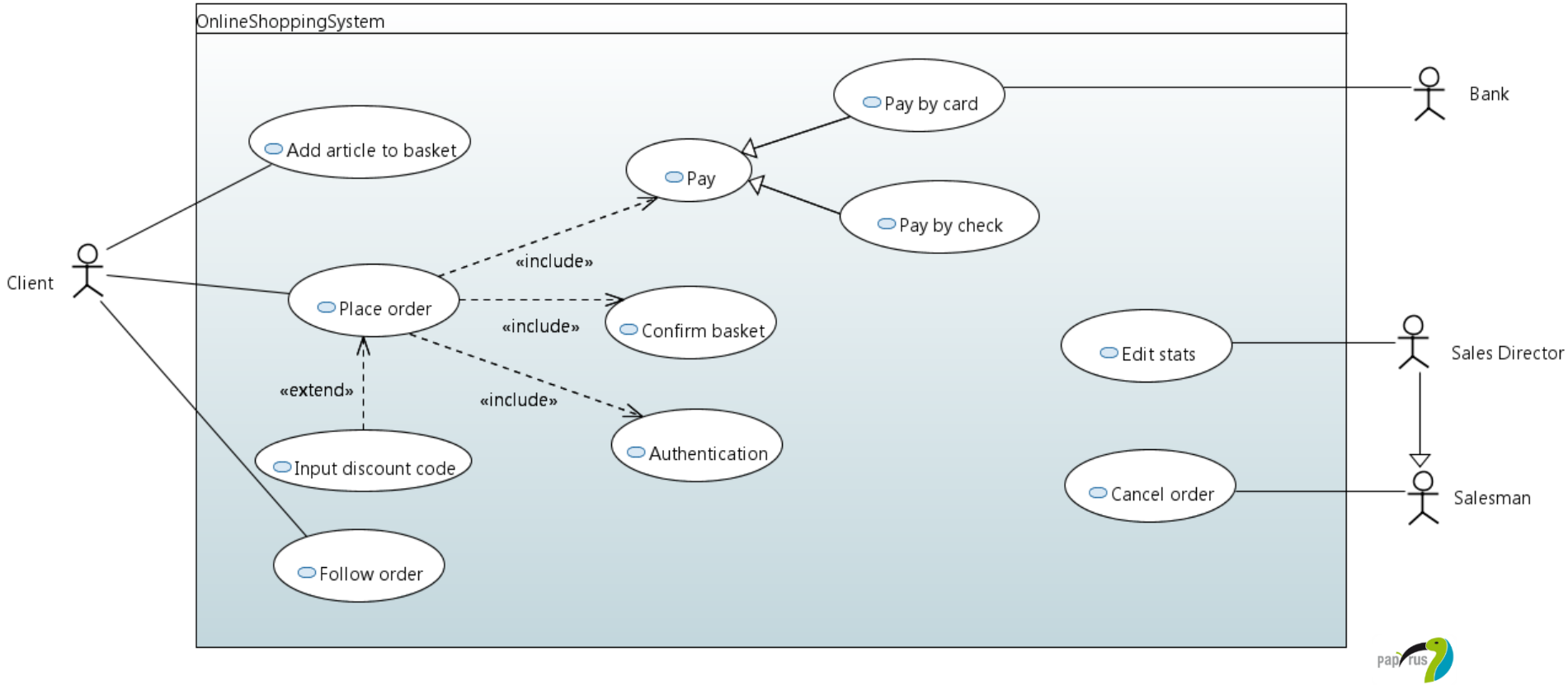
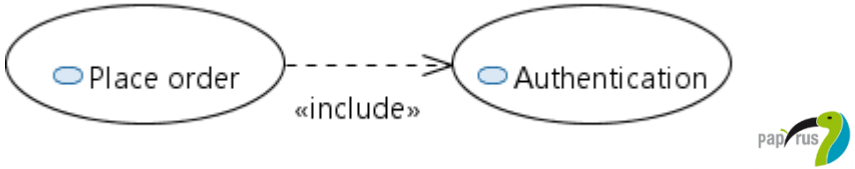
Example of complete use-case model

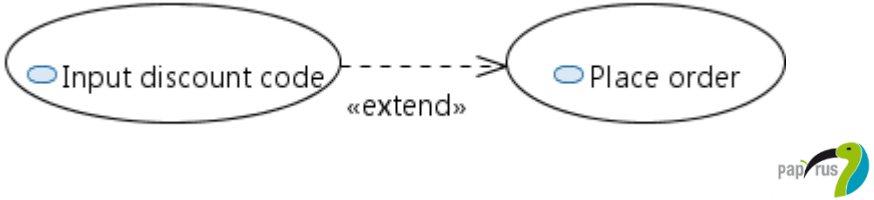


Inclusion and extension

- Inclusion: describes a use-case A is an required part of use-case B
- Example: “Place order” includes “Authentication” means the client must authenticate to place an order



- Extension: describes a use-case A is an optional part of use-case B
- Example: “Input discount code” extends “Place order” means the client may use a discount code when placing an order

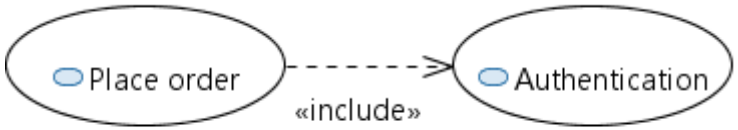


Inclusion and extension are dependencies

- **When use-case A includes or extends B, A depends on B**

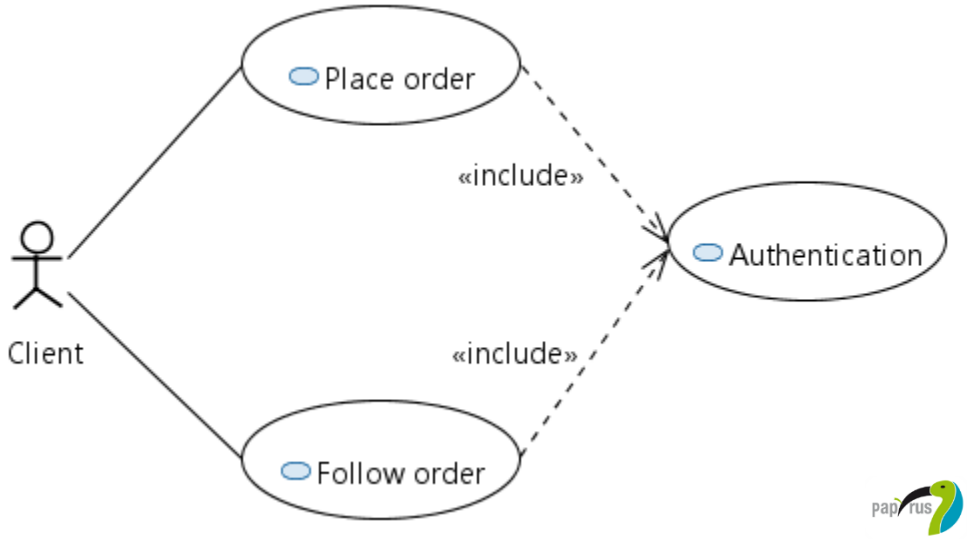
Reminder: When A depends on B, any modification to B may have an impact on A. In UML, dependencies are represented by dashed arrows.

- **Note on graphical notation:**
  - Since inclusion and extension are dependencies, they are share graphical notations with dependency, with keywords <<include>> and <<extend>> to distinguish them
  - Careful: the arrow indicates the direction of the dependency, not the direction of the inclusion relationship!
  - E.g. “Place order” depends on “Authentication” but it is “Authentication” that is included in “Place order”



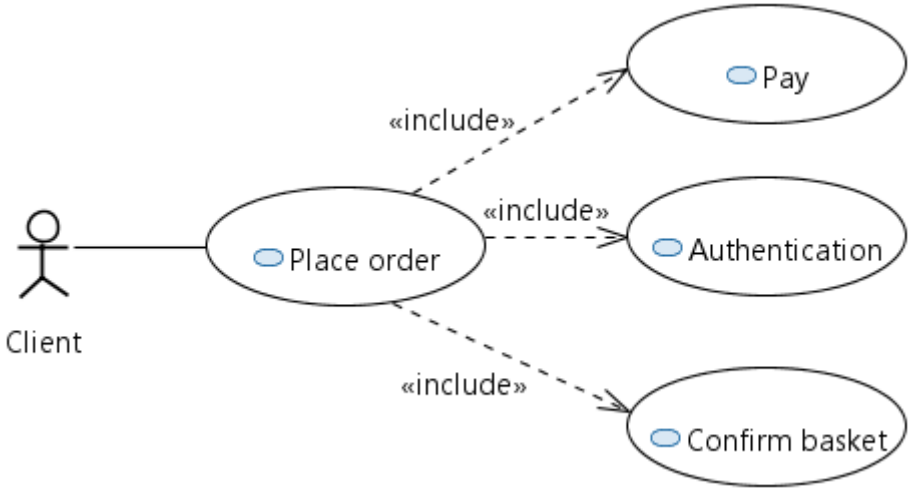
Re-usability through inclusions

- Use-cases can be re-used through inclusions
- Example: “Authentication” is re-used for both “Place order” and “Follow order” use-cases



Decomposition through inclusions

- Inclusion can decompose a complex use-case into several use-cases
- Example: “Place order” is a complex use-case that includes several simpler use-cases

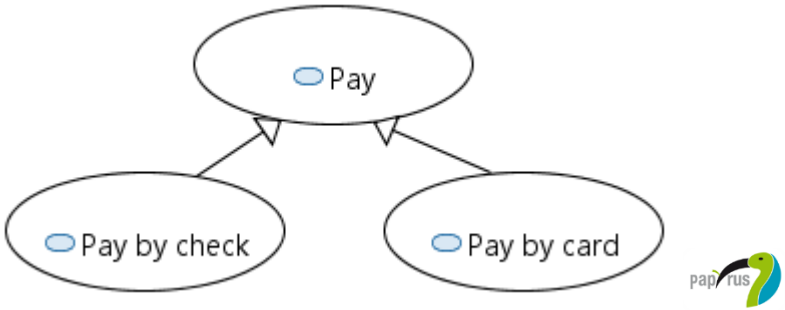


Inheritance between use-cases

- **A use-case can inherit another use-case**

Reminder: Inheritance = relationship between a general element A, and a specialized element B. When B inherits A, B “is a” A with specific features. Inheritance is modeled in UML with a generalization. Graphically it is an arrow with a hollow triangle pointing to the general element.

- **Example: “Pay by card” inherits “Pay” because paying by card is a special kind of payment; same for “Pay by check”**



Inheritance between actors

- The only relationship possible between actors is inheritance
- Example: A “Sales Director” is a “Salesman”

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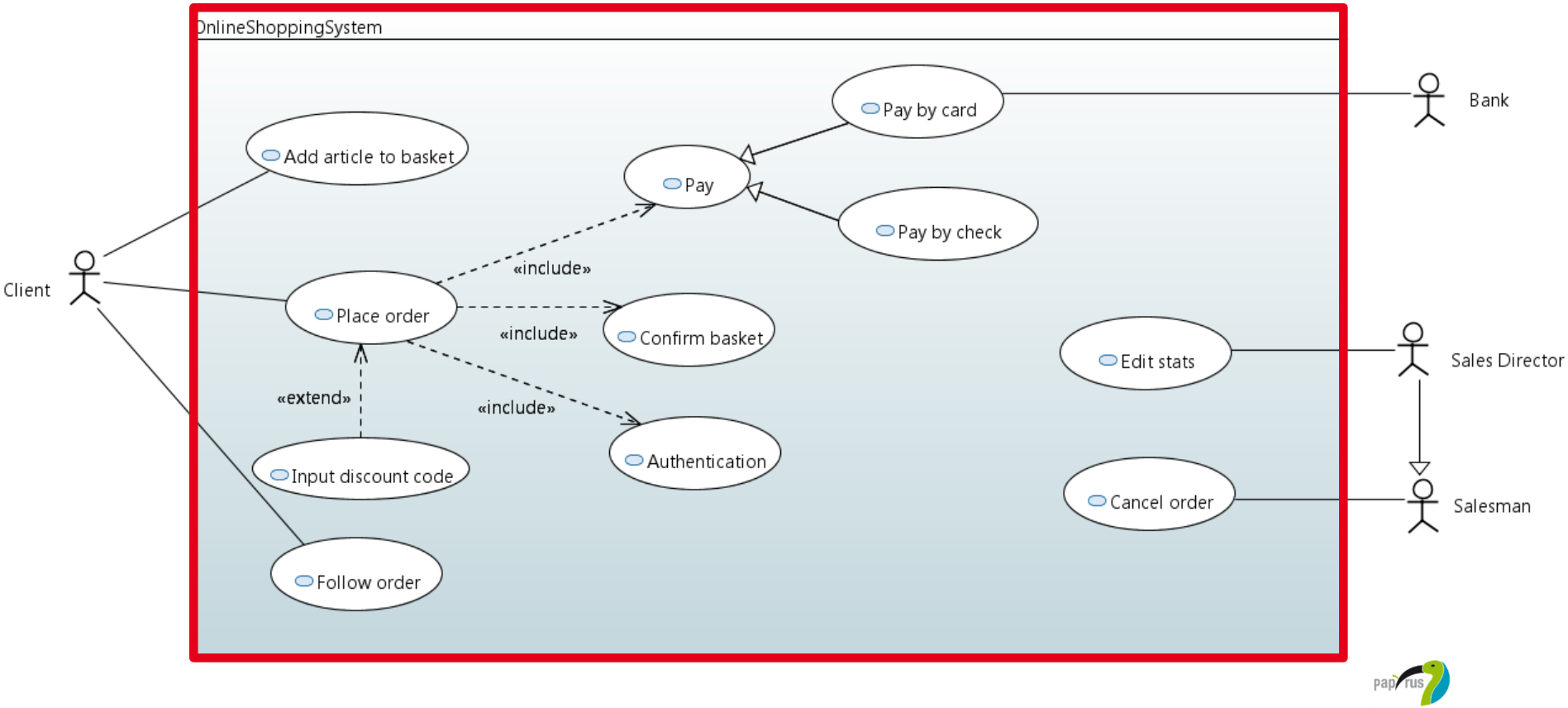
Identify actors

- **An actor is a user of the system**
- **An actor is a role, not necessarily a physical person:**
  - A same person can be represented as several actors if the person has several roles
  - If several persons play the same role relative to the system, they are represented as a same actor
- **Actors can also be, e.g.**
  - Devices used by the developed system (e.g. printer)
  - Available software that can be used by the developed system
  - Any external system that can interact with the developed system

Identify actors

- **To identify actors, think about the system border**



System border

Primary and secondary actors

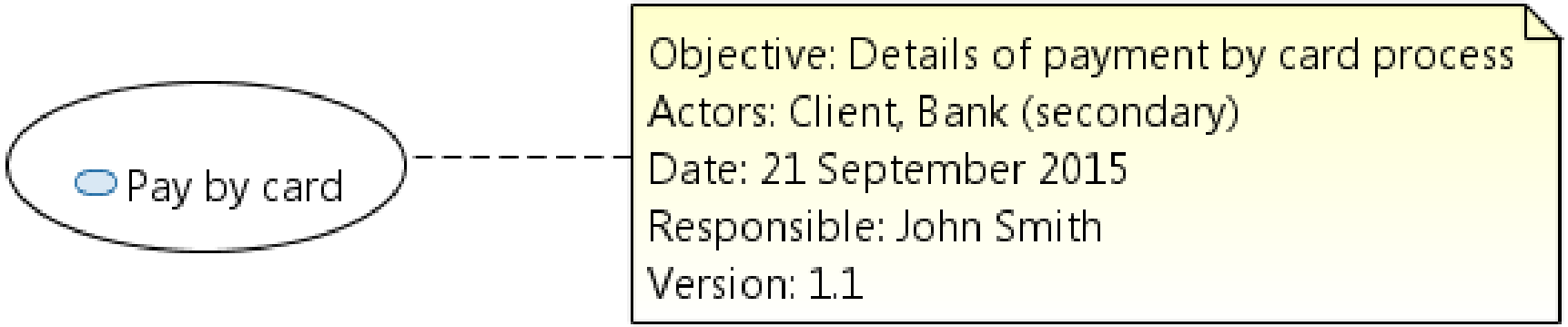
- An actor is said to be primary for a use-case, if the actor initiates the use-case
- A use-case may interact with a secondary actor that isn't the one that initiated the use-case
- E.g. secondary actors may be devices connected to the developed system
- Example: The "Client" initiates the use-case to "Pay by card" which interacts with the "Bank"; "Bank" is thus a secondary actor

Identify use-cases

- Identifying use-cases is not automatic
- Remember the subject is a black box: use-cases describe functionalities as seen by the outside world!
- Consider you are the actor and think of how you will use the system, in which cases do you use it, what are the services that you want to access
- Avoid redundancy and limit the number of use-cases by choosing the correct level of abstraction (e.g. several actions may be part of a same use-case)
- You are not modeling the details of each use-case, so stay at a level of abstraction that only describes the main services of the system
- Finding the correct level of abstraction is not simple, but you will become more experienced as you model use-cases

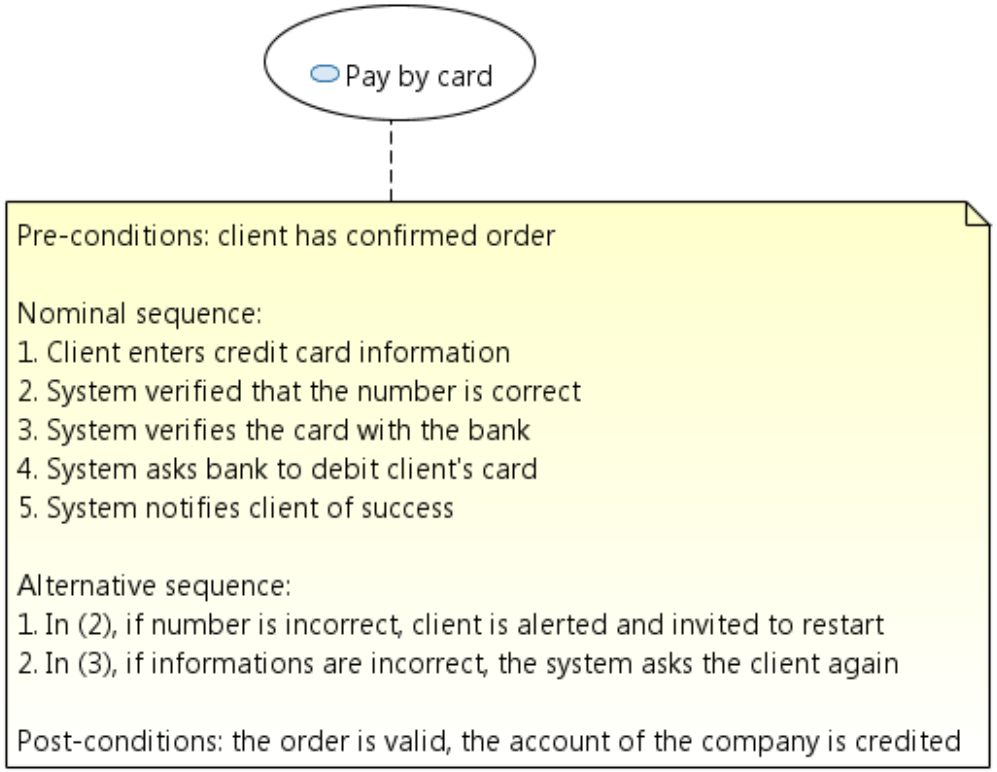
Describe use-cases

- **Too much detail in a use-case model is not good but too less is insufficient: a simple name for a use-case does not describe all aspects of a use-case**
- **The use-case must be described textually and the text may be in a UML comment**
- **Example:**



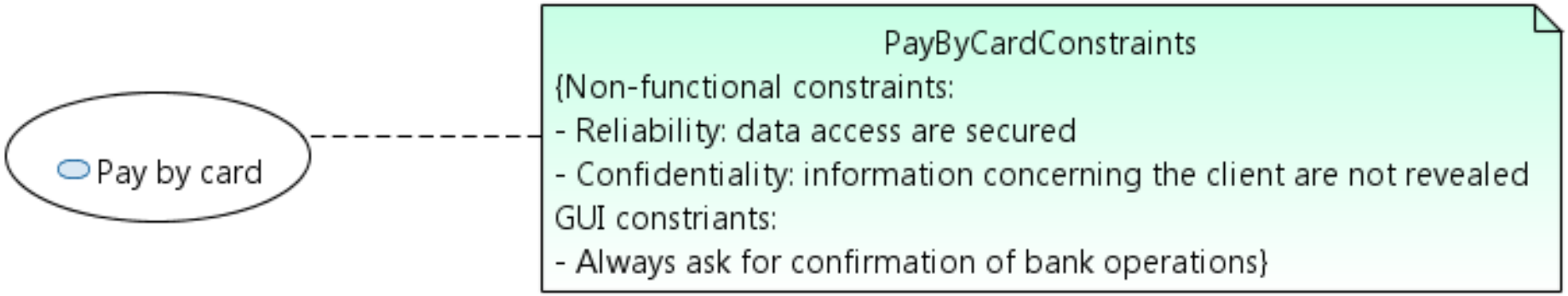
Describe use-cases

- You can also describe the behavior of a use-case
- Behavior can be modeled in UML as we will see in the future… for now let us consider a textual description only
- Example:



Describe use-cases

- **Finally you can also constrain a use-case**
- **UML constraints can be used for this purpose**
- **Example:**



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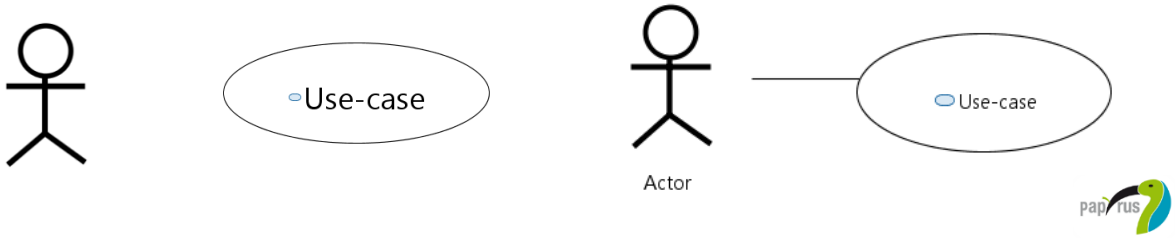
What are the basic elements of a use-case model?

!

Actor, use-case, and association actor-use-case



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What are the relationships between use-cases?

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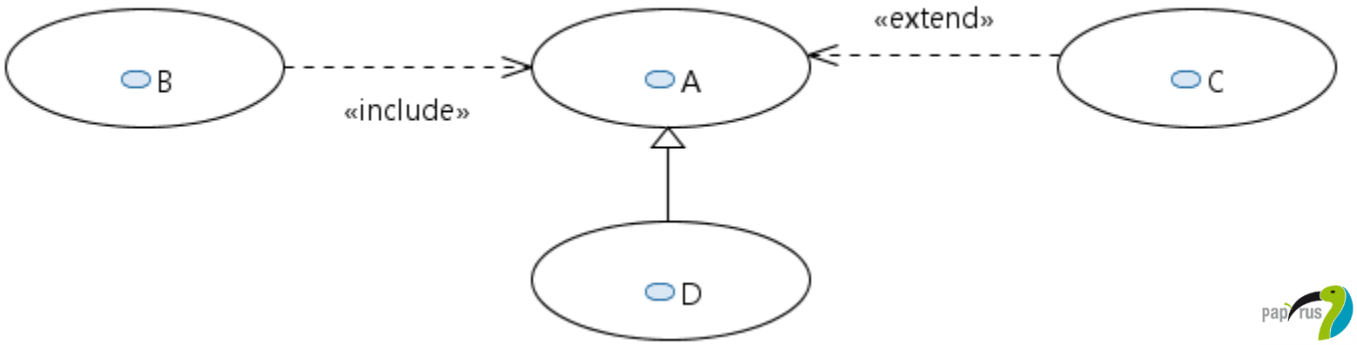
Inclusion, extension, generalization



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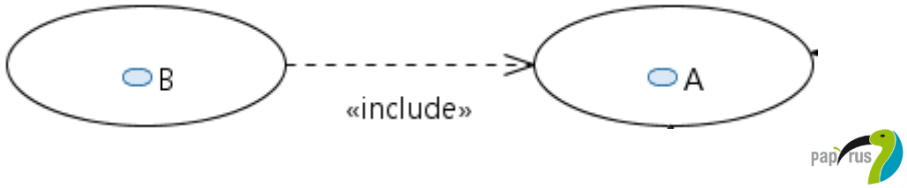
Use-case B includes A means A is:
(1) Optional (2) Required



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(2) Required

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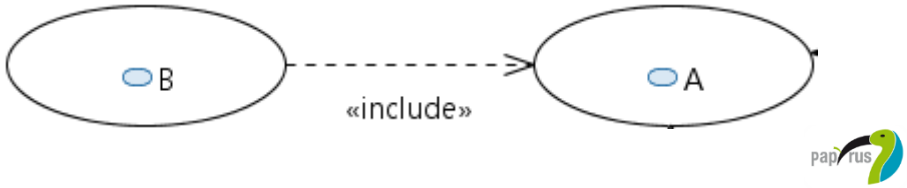
In the model below, who depends on who? Who is included in who?



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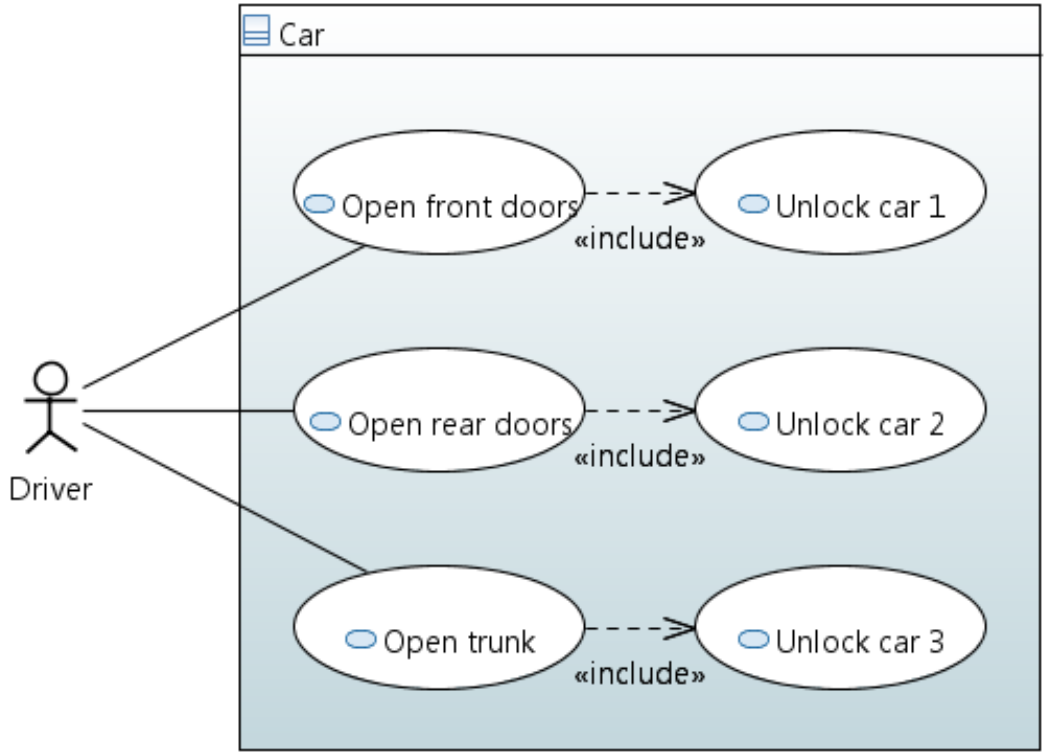
B depends on A, and A is included in B.

Is the model below optimal? If not, what is one way to optimize it?



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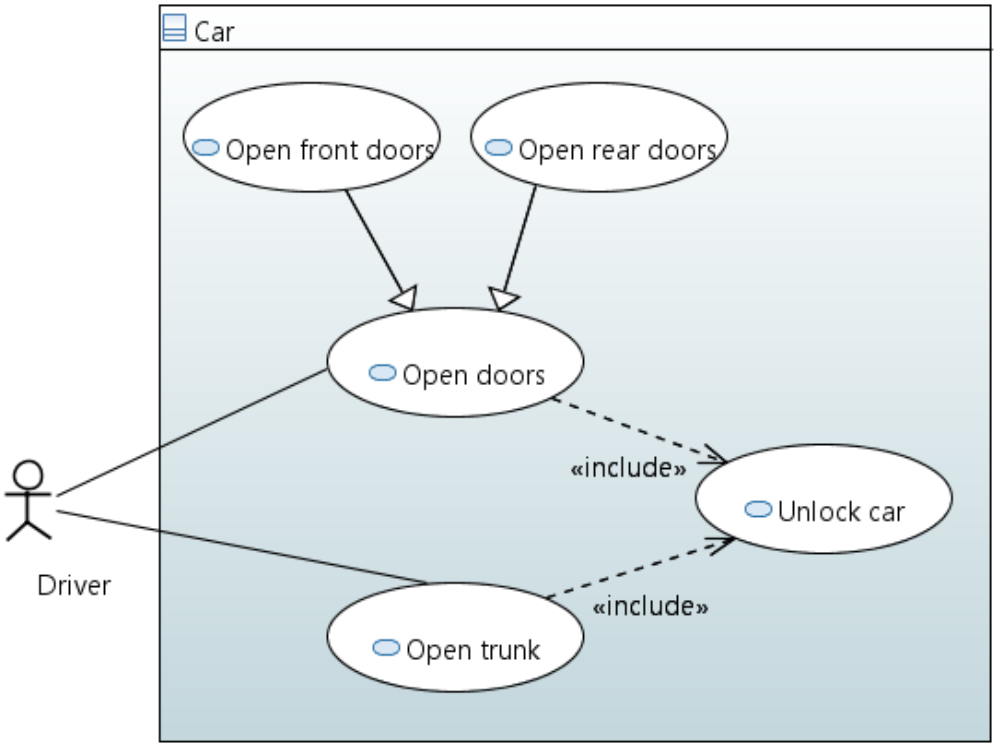
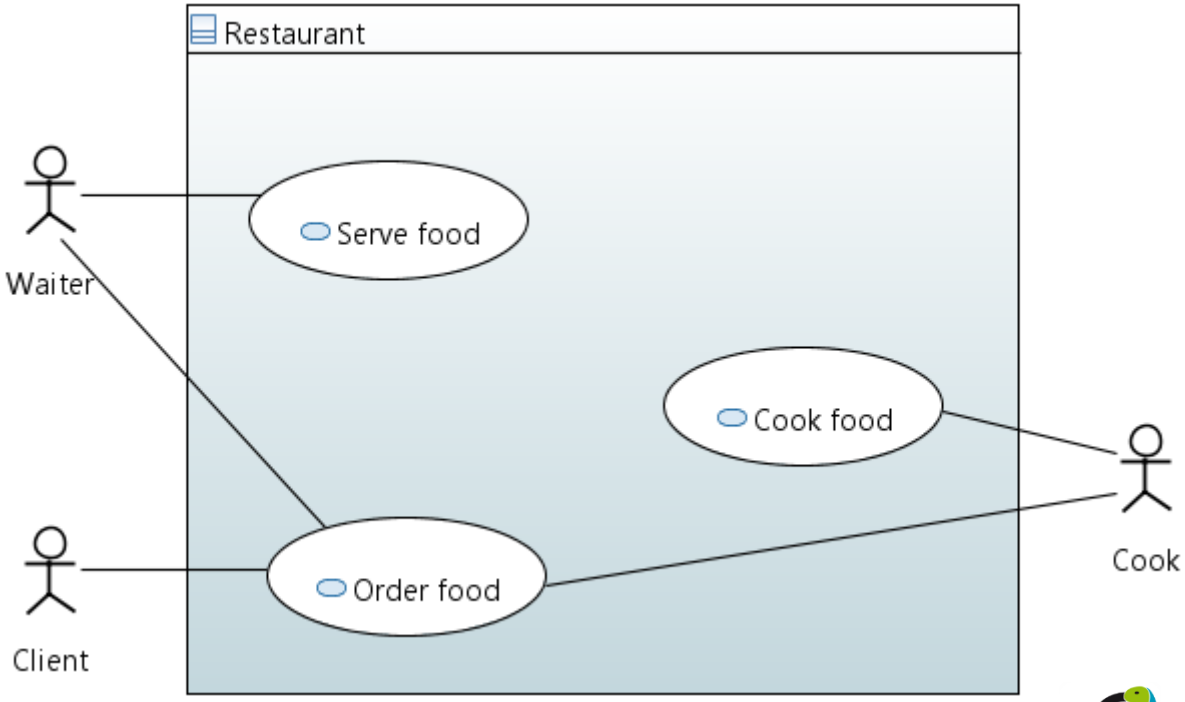
“Open front doors” and “Open rear doors” should be a generalization of “Open doors”. “Unlock car” should not be several use-cases, but one that is re-used.



Who are the secondary actors?



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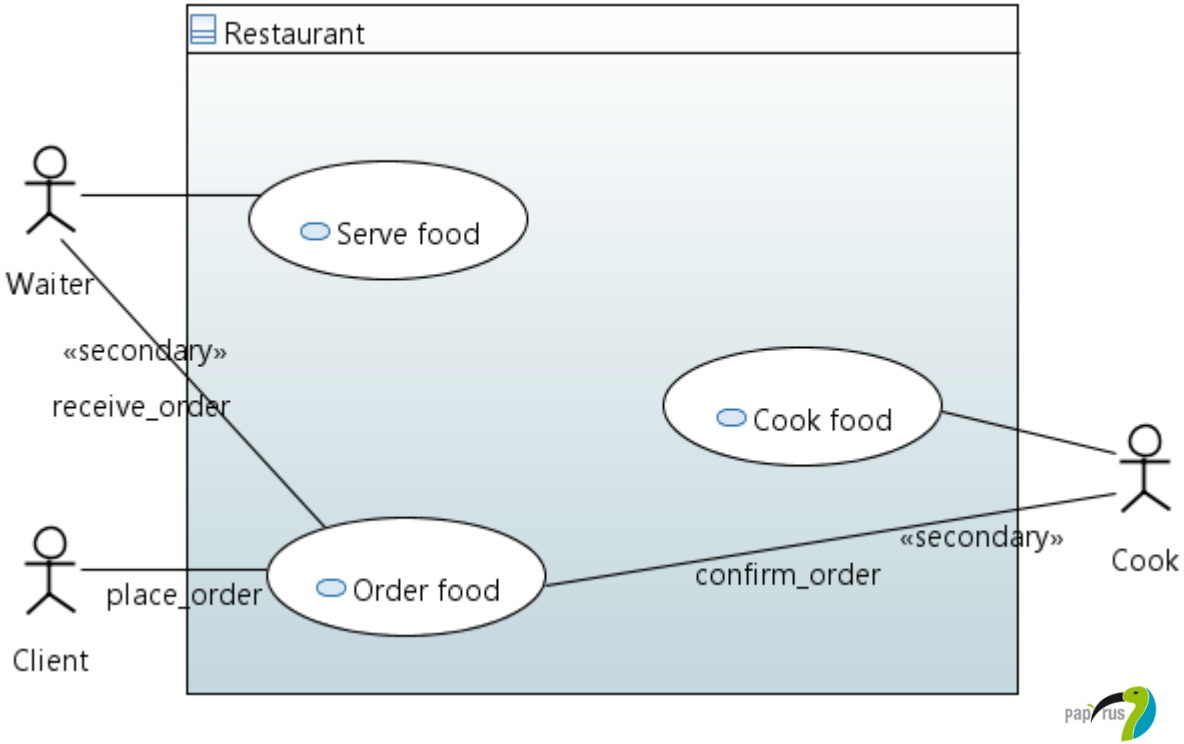
“Waiter” and “Cook” are secondary actors for “Order food” but they are also primary actors for “Serve food” and “Cook food” respectively.



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SUMMARY

- Before developing a system, it is important to analyze its requirements to show what it does, what needs it fulfills
- In UML, use-case models are used
- A use-case model has actors that interact with a system composed of use-cases
- Use-cases may be organized through inclusion, extension, and generalization relationships
- Actors are roles and they are identified by distinguishing the border of the system
- Use-cases must be modeled with correct level of abstraction…
- …but they must also be described, e.g. textually with UML comments and constraints